

Supplementary material

Machine Learning Models for Mental-Health Prediction from Actigraphy: A Systematic Appraisal and an Empirical Demonstration

Aminat Abiola Ajibola^{1,*}

Roger Nick Anaedevha²

¹College of Computer Science and Engineering, University of Hafr Al-Batin, Eastern Province, Kingdom of Saudi Arabia

²Institute of Cyber Intelligence Systems, National Research Nuclear University MEPhI, Moscow, Russian Federation

*aajibola@uhb.edu.sa Code and results: <https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal>

Contents

1 Table S1: TRIPOD+AI checklist	1
2 Table S2: feature definitions	2
3 Tables S3 and S4: identification sources and study list for Part I	3
4 Table S5: run manifests and prediction files	3
5 Figures S1 to S3: ROC and reliability curves	4

1 Table S1: TRIPOD+AI checklist

The checklist follows Collins et al. (BMJ 2024;385:e078378). Item numbering follows the extraction workbook used in Part I; section references are to the main article.

Table S1: TRIPOD+AI checklist (Collins et al. 2024) for this study; item wording abridged, numbering as in the companion review's extraction workbook.

Item	Requirement	Where reported
1	Title identifies development/validation of a prediction model, target population, outcome	Title
2	Abstract reports objective, data, population, outcome, predictors, sample size, model, performance	Abstract
3a	Background and rationale	Introduction para 1-2
3b	Objectives incl. development/validation	Introduction, final para (E1-E3)
4a	Data source described	Methods 2.1
4b	Study dates	Methods 2.1
5a	Setting, eligibility, recruitment	Methods 2.1; Table 1
5b	Treatments received	Methods 2.1 (not used)
6	Outcome definition and timing	Methods 2.2

Item	Requirement	Where reported
7	Predictors defined incl. measurement	Methods 2.3; Supplement S2
8	Sample size justified, EPV stated	Methods 2.4; Table 3 (EPV)
9	Missing data handling	Methods 2.5
10a	Pre-processing	Methods 2.5-2.6
10b	Model types and rationale	Methods 2.6
10c	Hyperparameter tuning	Methods 2.6 (fixed a priori)
10d	Internal validation (CV scheme, repeats, seeds)	Methods 2.7
10e	External validation method	Methods 2.8
11	Class imbalance handling; resampling inside folds	Methods 2.6
12	Performance measures incl. discrimination AND calibration	Methods 2.9
13	Model updating / recalibration	Methods 2.10
14	Fairness / subgroup performance	Methods 2.11; Results 3.5
15	Interpretability methods	Discussion (not an aim)
16	Funding	Declarations
17	Conflicts of interest	Declarations
18	Protocol / registration	Declarations (OSF)
19	Data availability	Data availability
20	Code availability	Code availability
21	Participant flow	Results 3.1
22	Participant characteristics incl. prevalence	Table 1
23	Full model specification	Methods 2.6; repository
24	Performance with uncertainty intervals	Tables 2-4
25	Subgroup / fairness results	Results 3.5; Table 5
26	Limitations	Discussion 4.3
27	Interpretation and implications	Discussion 4.4

2 Table S2: feature definitions

Table S2: Predictors computed for each valid calendar day of minute-level activity counts.

Feature	Definition
mean, sd, cv	Mean, standard deviation and coefficient of variation of counts over valid minutes
median, p10, p90, max	Median, 10th and 90th percentiles, maximum
skew, kurtosis	Sample skewness and excess kurtosis
prop_zero	Proportion of valid minutes with zero counts (inactivity)
mean_log1p, sd_log1p	Mean and SD of $\log(1 + x)$
day_mean, night_mean	Mean counts 08:00–22:00 and outside that interval
night_day_ratio	$\text{night_mean} / \text{day_mean}$
night_prop_zero	Proportion of zero minutes at night

Feature	Definition
M10, L5	Mean of the most active 10 h and least active 5 h of the hourly profile (circular)
relative_amplitude	$(M10 - L5)/(M10 + L5)$
IV	Intradaily variability of the hourly profile (Van Someren et al.)
m10_onset_sin, m10_onset_cos	Sine and cosine of the hour at which the M10 window starts (phase proxy)
acf_5, acf_60	Autocorrelation of the minute series at lags 5 and 60 min
psd_mean	$\log(1 + \text{mean power})$ of the within-day periodogram (excluding DC)
dominant_period_min	Period (min) of the dominant within-day spectral peak
n_transitions	Number of rest/activity transitions (zero to non-zero or back)
missing_frac	Fraction of the 1440 minutes without a valid count

3 Tables S3 and S4: identification sources and study list for Part I

Table S3: Supplementary Table S3. Identification sources for Part I (September 2026).

Source	Query / procedure
Google Scholar	“Depresjon” OR “Psykose” OR “Hyperaktiv” OR “OBF-Psychi
Semantic Scholar	citation graph of the four dataset papers
PubMed / PMC	Depresjon[tiab] OR Psykose[tiab] OR Hyperaktiv[tiab] OR “OB
arXiv / medRxiv / TechRxiv	full-text search for the four dataset names
Publisher sites (Springer, Elsevier, MDPI, ACM, IEEE, JMIR)	full-text search for the four dataset names

Table S4: Supplementary Table S4. Study sources and extraction basis.

ID	Source	Extraction basis
S01	doi:10.1145/3204949.3208125	full text
S02	Feature extraction in motor activity signal (Diagnostics 2019)	abstract/metadata
S03	doi:10.1145/3347444.3356238	abstract/metadata
S04	doi:10.1371/journal.pone.0231995	abstract/metadata
S05	doi:10.1109/CBMS49503.2020.00064	full text
S06	doi:10.1145/3458305.3478454	full text
S07	PMC9495338	abstract/metadata
S08	PMC9318635	abstract/metadata
S09	ResearchGate 362753700	abstract/metadata
S10	arXiv:2303.07847	abstract/metadata
S11	arXiv:2310.09277	abstract/metadata
S12	PMC10377958	abstract/metadata
S13	SciELO S0188-95322023000400038	abstract/metadata
S14	doi:10.1007/s11042-023-17119-6	abstract/metadata
S15	doi:10.1007/s00500-024-09832-7	abstract/metadata
S16	doi:10.1016/j.bspc.2024.105907 (verify)	abstract/metadata
S17	doi:10.1007/s11042-024-19631-9	abstract/metadata
S18	doi:10.1007/s41870-024-01895-x	abstract/metadata

ID	Source	Extraction basis
S19	doi:10.2196/72038	abstract/metadata
S20	doi:10.1038/s41598-025-07245-6	abstract/metadata
S21	PMID 40588896	abstract/metadata
S22	doi:10.1007/978-3-032-05507-1_16	abstract/metadata
S23	doi:10.36227/techrxiv.176403977.77890479	abstract/metadata
S24	doi:10.1038/s41597-025-04384-3	full text
S25	doi:10.1101/2025.08.26.25332257	abstract/metadata
S26	arXiv:2512.00103	abstract/metadata
S27	doi:10.1007/s44163-026-01574-9	abstract/metadata
S28	PMC9312518	abstract/metadata

4 Table S5: run manifests and prediction files

Every experiment run writes `results/real/<experiment>/manifest.json` with the fields in Table S5, and every model-splitter combination writes its out-of-fold predictions to `results/real/<experiment>/predictions/<cohort>.<model>.<splitter>.csv` with one row per seed, fold and day window (columns: seed, fold, window_id, subject_id, cohort, y, p). The tables in the main article are regenerated from these files by `python-mmlmhretable` and `scripts/update_readme.py`; the Part I figure and table by `scripts/part1_figure.py`; the calibration and radar figures and the comparison table by `scripts/part2_extra_figures.py`.

Table S5: Fields of a run manifest.

Field	Content
created_utc	Timestamp of the run
git_sha, git_dirty	Commit of the code that produced the run; whether the tree had uncommitted changes
config_hash, config	SHA-256 prefix of the resolved configuration and the configuration itself (seeds, folds, mode)
python, platform, package_versions	Interpreter, OS, and versions of numpy, pandas, scikit-learn, xgboost, imbalanced-learn, sci
dataset_checksums	SHA-256 of every raw CSV file of every cohort used
synthetic	Whether the run used the synthetic fixture (always false for reported results)
experiment, n_rows	Experiment name and number of result rows written

5 Figures S1 to S3: ROC and reliability curves

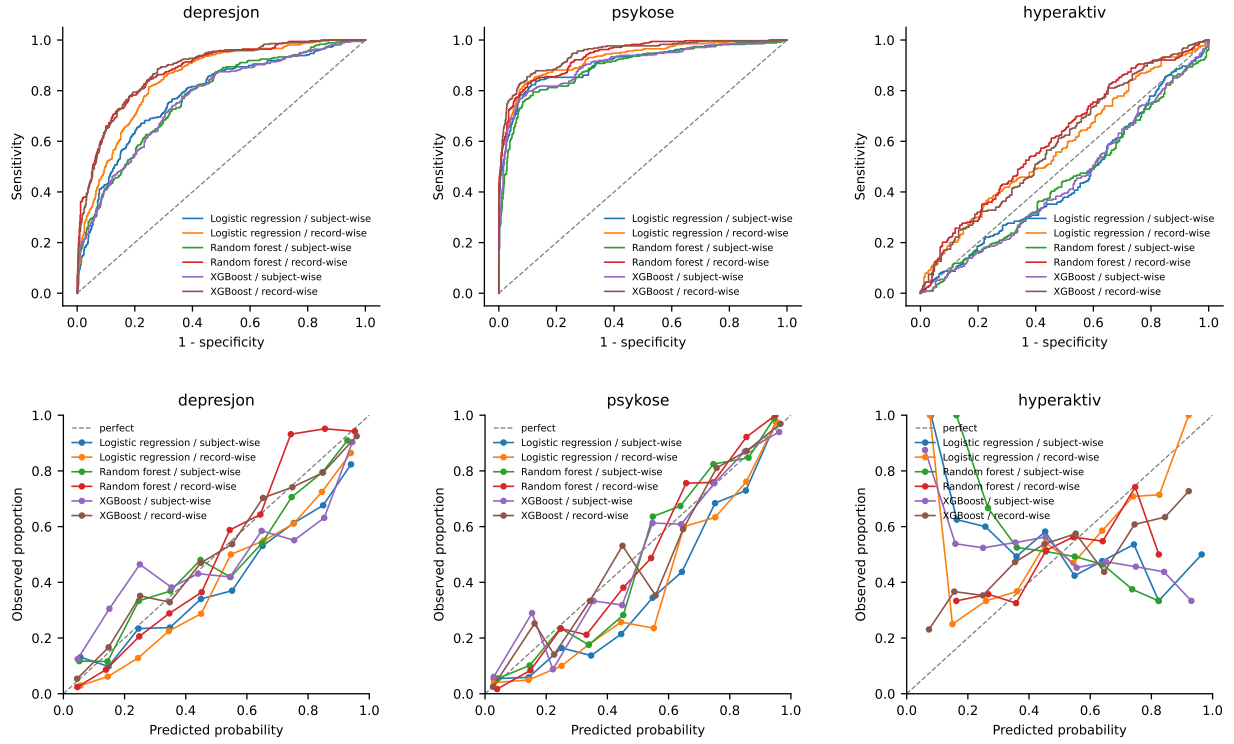


Figure S1: E1. ROC (top) and reliability (bottom) curves of the seed-averaged out-of-fold predictions for every model under subject-wise and record-wise cross-validation, per cohort.

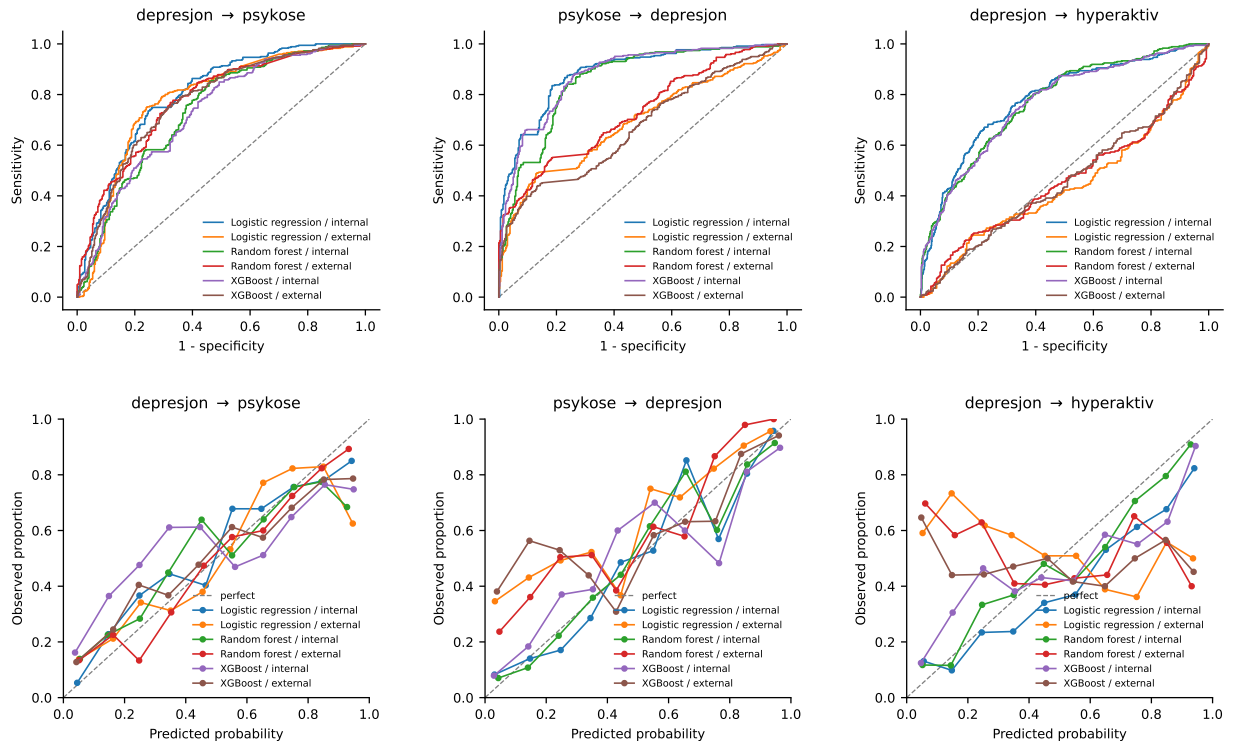


Figure S2: E2. ROC (top) and reliability (bottom) curves of internal (subject-wise CV on the training cohort) and external (frozen model on the test cohort) predictions for each cohort pair.

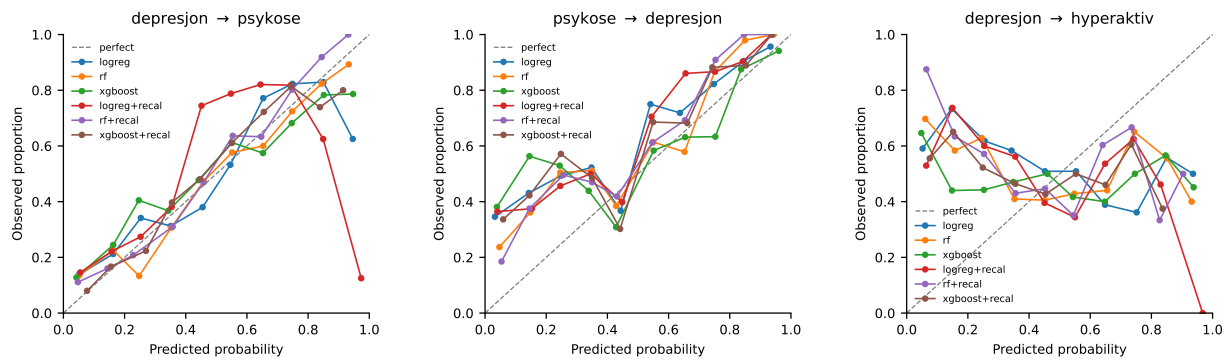


Figure S3: E3. Reliability curves of external predictions before and after logistic recalibration fitted on the training cohort's out-of-fold predictions, per cohort pair.